Core Evidence 2: ML Workflow Diagram

IT3091 Machine Learning — Industry Explorer Track

Group ID: `2026-AI-36`

Version: Final Submission — October 2026

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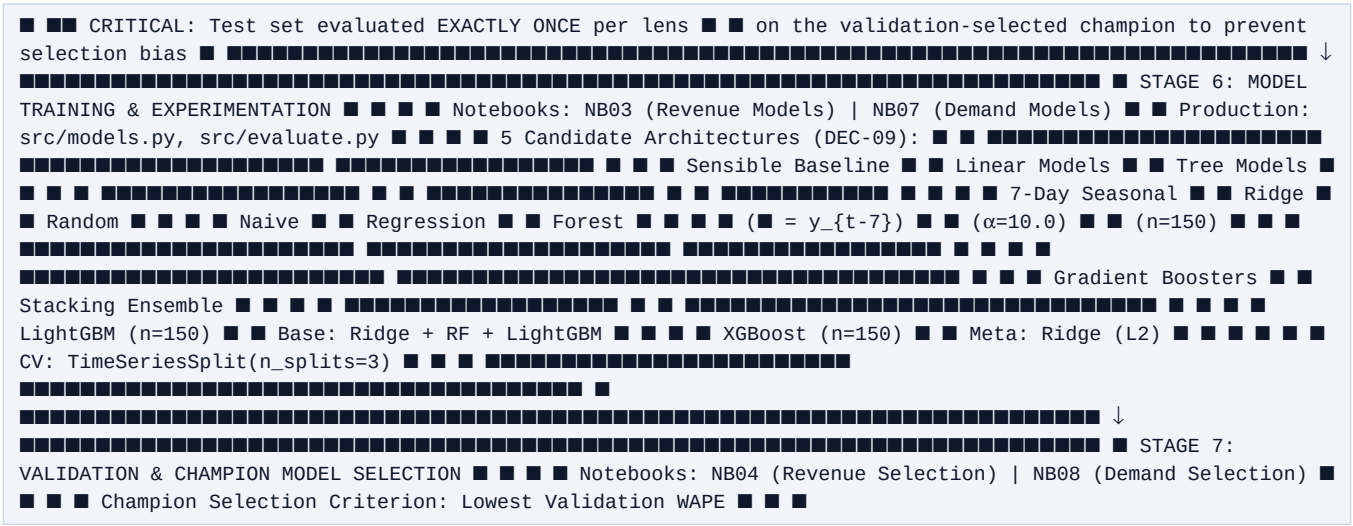
1. End-to-End Machine Learning Workflow: ChefCast AI

The following narrative describes the complete ML workflow implemented in this project, following the structured, evidence-based process required by the IT3091 assessment rubric.

```
■■■
```


■ ■■ CRITICAL: Test set evaluated EXACTLY ONCE per lens ■ ■ on the validation-selected champion to prevent selection bias ■ ↓ ■ STAGE 6: MODEL TRAINING & EXPERIMENTATION ■ ■ ■ ■ Notebooks: NB03 (Revenue Models) | NB07 (Demand Models) ■ ■ Production: src/models.py, src/evaluate.py ■ ■ ■ ■ 5 Candidate Architectures (DEC-09): ■ ■ ■ ■ ■ Sensible Baseline ■ ■ Linear Models ■ ■ Tree Models ■ ■ ■ ■ ■

EDA (Lens 2)	NB05	IT24102726	Demand EDA, product selection
Feature Eng. (Lens 2)	NB06	IT24102913	41-feature Lens 2 panel matrix
Modeling (Lens 2)	NB07	IT24102572	5 candidate model experiments
Validation (Lens 2)	NB08	IT24102664	Champion: Ridge Regression
Benchmark Synthesis	NB09	Joint	Cross-lens comparison & Ensemble

3. Data Flow Summary

```
pos_raw_dataset.csv (127,093 rows) ↓ [data_pipeline.py: clean, filter, aggregate]
cleaned_pos_transactions.csv (125,912 rows) ↓ [group by date / group by date+item] daily_revenue_series.csv
daily_product_demand_long.csv (895 days × revenue) (9,845 item-days × 11 dishes) ↓ [feature_engineering.py]
Lens 1 Feature Matrix Lens 2 Feature Matrix (874 days × 30 features) (9,614 item-days × 41 features) ↓
[evaluate.py: train/val/test split + 5 models] models/metrics.json models/lens*.joblib (×12 files) ↓
[forecasting.py: recursive forward engine] 7-day forward forecast arrays ↓ [app.py: Streamlit UI] Interactive
Decision Support Dashboard (http://localhost:8501)
```